

BBO capstone project presentation template

Develop a concise presentation that captures:

1. An overview of your BBO approach

- Describe what you are trying to achieve or discover through the BBO capstone project. Focus on the core objective of your optimisation process, using clear and accessible language.
- Briefly describe your overall strategy and how your process fits together.

The central objective of the project was to find an input vector X of dimensions (2D-8D) that would maximize the output of eight black-box functions. This objective was achieved by applying a learning-based progressive analysis of results from each iteration, within a limited evaluation budget strategy. The problem was approached by managing the exploration-exploitation trade-off. First, exploration was performed in the function's value space to discover regions where a maximum value could be located. Subsequently, exploitation of these regions was initiated to find the maximum value within each region.

The optimization strategy was structured as an iterative system with progressive information accumulation and modular integration. During the initial iterations, exploration was primarily conducted using surrogate GP models, EI/UCB acquisition functions, and the HEBO model to gain a general understanding of the function. Once more information was accumulated through the iterations, patterns began to emerge, suggesting the formation of clusters with high values. For this reason, k-NN models were integrated to identify neighborhoods, clustering to identify regions with potential optimal values, and SVM and NN classification models to distinguish between high- and low-performance zones. All of this was done to gain a deeper understanding of each function's structure. Ultimately, the optimization strategy evolved into a system that maintains a cycle of proposing, evaluating, interpreting, and adjusting a point in each iteration, seeking to maximize the function.

2. How your strategy has evolved

- Describe the key changes in your approach since your early rounds.
- What influenced these changes – data trends, feedback, model performance or intuition?
- Identify which principles or heuristics now guide your query decisions.

2.- The optimization strategy evolved from an exploratory approach focused on the results obtained by the optimization models (GP+EI, GP+UCB) to a much more structural approach to the function.

In the first iterations, exploration was prioritized using only Bayesian Optimization (BO) to identify regions where the function's maxima could be located. However, with each iteration, more information was obtained, and patterns began to emerge in the results, mainly compact clusters of maximum values and clear separations between high- and low-value regions. Therefore, additional layers of analysis, such as clustering, SVM, and Neural Networks (NN), were used to gain a much deeper structural understanding of the functions in subsequent iterations. The changes described were primarily driven by trends in the data, which, although captured by the BO models, could be better represented by the complementary models.

All this integration of models in an iterative system allowed the standardization of the decision-making process guided by clear heuristics, where exploitation will be carried out as long as there is evidence that shows that improvements in the maximum value are achieved within the cluster or that there are possible improvements. Exploration will be applied when a promising region has not been located or when marginal improvements are obtained within the cluster.

3. Patterns, data and insights

- Highlight the most meaningful trends you've observed across your data points or model evaluations.
- Identify which variables or behaviours seem to influence results the most.
- Explain how these observations have shaped your understanding of the search process.

3.- One of the most relevant patterns within the strategy was the clustering of regions with high values within the space of each function. As more information became available, the clusters could be classified into those that were already well-established, where each evaluation showed diminishing returns; those that were not yet established but still showed improvements with each iteration; and those that were very diffuse, as they were widely dispersed with distant neighbors. Therefore, the candidates began to be evaluated not only on their point value but also on their position within the cluster and the proximity of their neighbors.

Regarding the influential variables, although a PCA analysis was not performed to determine the most influential variables in the evaluation definitively, it was possible to identify, in some functions, variables that produced significant changes in the output with minimal modifications to the input, as well as others where modifications resulted in only marginal changes in the evaluation. A very clear example in my case was function F8, where, during four iterations, changes to dimensions 4 and 7 produced significant improvements in optimization. In contrast, changes to the other dimensions did not generate any additional gain.

This behavior changed my understanding of the process. I began to think of optimization not only as the estimation of new optimal points, but also as a set of factors that directly influence the search for an optimal value, such as the patterns and variables that most strongly influence the function's behavior.

4. Decision-making and iteration

- Explain how you balance exploration and exploitation.
- Describe one or two examples of strategic decisions. What worked, what didn't, and why?
- Reflect briefly on how you handle uncertainty or adapt when results don't match expectations.

4.- The balance between exploration and exploitation evolved dynamically as more information accumulated. Initially, a broad exploration was conducted to reduce uncertainty in the function's value space. Subsequently, when patterns were identified, mainly compact clusters with a concentration of high values, exploitation focused on those regions began, attempting to find an optimal value within that region. During these later iterations, I have opted for exploration when I identify dispersion or diminishing returns, and have continued with exploitation as long as I see continued improvements in evaluations in a given region.

As an example, I would like to mention the overall result of the optimization strategy I designed, based on the balance described between exploration and exploitation. During the initial iterations, exploration helped me gain a general overview of each function's structure. However, once promising clusters were identified, I began exploitation without considering a maximum number of iterations per region before returning to exploration, leading my strategy to overexploit the identified clusters in most functions, creating a trade-off between overexploitation and exploration to identify potentially more promising regions. I believe the strategy can be improved and didn't completely fail; I think that, under the assumptions I built it on, it only applies when the function is smooth and has few local maxima.

Usually, when the results don't match expectations, I don't discard the model's estimates; instead, I give them less weight in the decision-making process. In those cases, I give greater weight to elements that provide information about the function's overall structure. This practice has helped me manage uncertainty without deviating from the optimization strategy.

5. Next steps and reflection

- Outline your planned actions or experiments to improve the model's performance.
- Include two or three sentences connecting your work to the broader ML landscape.
- Add a short statement on how you'd communicate your results to a non-technical audience or stakeholder.

5.- As next steps, I would focus primarily on a couple of actions. First, I would define a stopping criterion based on metrics (e.g., cluster dispersion, neighborhood, etc.) to avoid overexploiting a single cluster and encourage exploration. Second, I will fine-tune the HEBO model, as it is one of the most powerful tools, and a better understanding of the model will allow for estimates that facilitate determining the optimum. Finally, I would develop a more adaptive RL-type approach that dynamically adjusts the balance between exploration and exploitation based on the system's state.

This work is fully aligned with current trends in machine learning, as it enables integrating models such as Bayesian Optimization, k-NN, Clustering, SVMs, NNs, and RL to efficiently estimate the global maximum of a function with limited data while making decisions under uncertainty. Furthermore, it demonstrates the cohesion of hybrid systems that combine models with human interpretation.

The way I would communicate the results to a non-technical audience would be as follows: "We developed a decision-making system that allows the optimal value of a function to be identified efficiently and iteratively in an environment of restricted evaluations and under high uncertainty."